EXP NO : 3

Date :

DETERMINATION OF VOLTAGE IN CIRCUIT USING NODAL ANALYSIS :

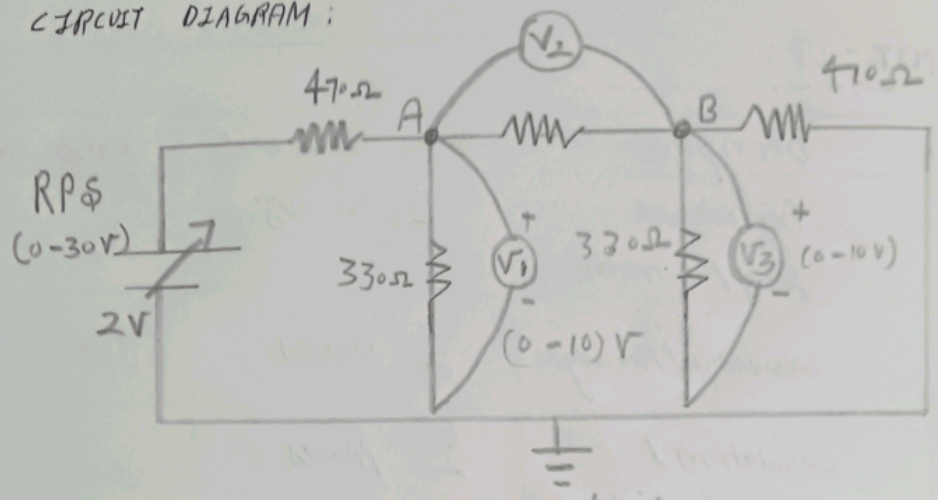
AIM :

To determine the voltage in the circuit using nodal analysis both theoretically & practically for a given DC circuit

Apparatus Required :

S. No	Apparatus	Specification	Quantity
1.	Regulated power Supply (RPS)	(0-30V)	1
2.	Multimeter	---	1
3.	Resistors	470Ω, 330Ω	3, 2
4.	Bread board	---	1

CIRCUIT DIAGRAM :



Nodal circuit

TABULAR COLUMN :

parameters	theoretical	Practical
Supply voltage	2V	2V
V_A	0.64V	0.49
V_B	0.19V	0.35
V_{AB}	0.45V	0.14

CALCULATIONS :

Node A:

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$113 V_A + 33 V_B = 66 \rightarrow (1)$$

Node B:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$113 V_A - 33 V_B = 0 \rightarrow (2)$$

From (1) & (2)

$V_B = 0.19 V$

$V_A = 0.64 V$

PROCEDURE :

1) Give connections as per circuit diagram
2) Switch on the supply vary the RPS
3) Note down the readings of ammeters & voltmeters and tabulate them
4) vary the RPS for ~~additional~~ different input voltages and note down the readings of all the meters
5) Reduce the RPS to its minimum value & switch off the supply.
6) using the tabulated values, verify Kirchoff's laws practically and verify it theoretically.

MARKS SPLIT - UP :

S. No	Description	Marks Allocated	Marks obtained
1.	Experimental setup/connections	20 Marks	20
2.	observation / Readings	20 Marks	20
3.	Calculations / Results	20 Marks	20
4.	Viva	20 Marks	10
5.	Record	20 Marks	10
		Total	80

RESULT :

Thus the Nodal analysis is verified practically and theoretically, the resultant voltages for 2 V supply are :

a) The voltage V_1 is 0.49 V

b) The voltage V_2 is 0.35 V

c) The Voltage V_3 is 0.14 V

80